

Shape optimisation of nonlinear Naghdi shells on discrete geometries

Ado Farsi^{a,b,c,*}, Alberto Paganini^d

^aImperial College London, Department of Earth Science and Engineering, London, United Kingdom

^bUniversity College London, Department of Earth Sciences, London, United Kingdom

^cTanuki Technologies, London, United Kingdom

^dUniversity of Leicester, School of Computing and Mathematical Sciences, Leicester, United Kingdom

Abstract

A thin shell carries load through its shape and, at finite deflections, its stiffness changes with the load itself, so a shape optimum found with a linear model can be far from optimal. We present an automated framework that embeds the fully geometrically nonlinear shell response in the shape-optimisation loop. The forward model, a five-parameter nonlinear Naghdi shell stabilised against locking by partial selective reduced integration, operates directly on a discrete (faceted) triangulation with a numerically recovered director field — dispensing with the exact mid-surface parameterisation of isogeometric approaches, a chart that ceases to exist once the geometry itself is the design variable. Implemented in Firedrake, the model generates its residual, consistent tangent and adjoint automatically; shape derivatives, computed by algorithmic differentiation through the full load-continuation solve, drive the Fireshape/ROL trust-region optimiser. The forward solver reproduces the Sze/Abaqus benchmark for a clamped semi-cylindrical shell under a point load, capturing the progressive stiffening that a geometrically linear model cannot reproduce. The optimisation is validated against the COMSOL benchmark, a sheet-metal bracket under bending: the framework develops the same off-mid-plane corrugation mechanism and attains an 87 % reduction of elastic strain energy *within* the prescribed displacement budget, matching the benchmark’s magnitude and area change. Applied to the curved semi-cylinder, it forms a smooth stiffening crease that reduces the shell’s average deflection under load by 78 %.

Keywords: nonlinear Naghdi shell, shape optimisation, Firedrake, Fireshape, adjoint method, PDE-constrained optimisation, PSRI

Highlights

- Nonlinear Naghdi shell on faceted meshes with a numerically recovered director
- Geometry is the design variable: no analytic mid-surface chart required
- Adjoint shape derivatives by algorithmic differentiation through load continuation
- Validated on the Sze/Abaqus forward and a COMSOL shell optimisation benchmark
- Multilevel control, smoothing metric and enforced symmetry give large stiffening

1. Introduction

Thin-walled shells are ubiquitous in lightweight engineering: a curved sheet of material carries load predominantly through membrane action and so attains a high stiffness-to-weight ratio. Because performance depends so strongly on curvature, the *shape* of a shell is a powerful design lever, and shape optimisation is a natural route to lighter, stiffer components. When the operating loads are large, however, the structural response is geometrically nonlinear: finite

*Corresponding author.

Email addresses: `ado.farsi@imperial.ac.uk` (Ado Farsi), `ado.farsi@tanuki.ai` (Ado Farsi), `a.paganini@leicester.ac.uk` (Alberto Paganini)

rotations, membrane–bending coupling and load-dependent stiffening (or softening, up to snap-through) all matter, and an optimum identified on a linearised model may be far from optimal — or even unsafe — once the full nonlinear response is accounted for. A predictive shape-optimisation framework for shells must therefore embed a geometrically nonlinear forward model in the optimisation loop.

Two developments make this practical. First, automated finite-element environments based on a symbolic weak-form language, such as Firedrake [1] and the Unified Form Language [2], generate the residual, the consistent tangent and, through algorithmic differentiation [3], the adjoint directly from an energy functional. This removes the hand-coding that has historically made nonlinear shell elements and their sensitivities laborious and error-prone; the nonlinear Naghdi shell of Hale et al. [4] is a compelling demonstration. Second, shape-optimisation toolboxes such as Fireshape [5] provide a rigorous diffeomorphism-based control space and Sobolev descent metrics that mitigate mesh-dependence and promote mesh quality, so that the domain itself can be treated as the optimisation variable with virtually no additional user code or computational cost.

Shell shape optimisation has a long history [6, 7], and the contemporary literature is dominated by isogeometric analysis (IGA), in which Kirchhoff–Love shells are discretised on the same NURBS/spline basis used to describe the CAD geometry [8, 9, 10, 11, 12], with applications ranging from aerostructures to wind-turbine blades [13]. The recent open-source GOLDFISH framework [14] couples isogeometric Kirchhoff–Love shells to a general optimisation back-end and is the closest analogue to the present work. A common thread of the IGA approach is that it relies on an *exact, closed-form* parameterisation of the mid-surface: the geometry is a smooth spline map from which the metric, curvature and normal fields are evaluated analytically.

This reliance on an analytic chart is precisely what becomes problematic when the mid-surface is the unknown. In a general finite-element framework the shell is a faceted triangulation with no closed-form parameterisation, and every shape update produces a new mesh. The contribution of this paper is a framework that removes this obstacle and validates it end to end: (i) a geometrically nonlinear, five-parameter Naghdi shell formulated on a *discrete* (faceted) mesh, in which the director field, reference metric and curvature are recovered numerically from the triangulation rather than from an analytic map (Sections 2–3); (ii) an adjoint-consistent coupling of this nonlinear shell to the Fireshape diffeomorphism control space, yielding shape derivatives entirely by algorithmic differentiation (Section 4); and (iii) validation of the forward model against the Sze/Abaqus benchmark [15], validation of the shape optimisation against the COMSOL “Shape Optimization of a Shell” benchmark [16] on an industrially relevant sheet-metal bracket, and a shape-optimisation showcase on the curved semi-cylinder used for the forward validation (Section 5). The entire forward and adjoint model is expressed in fewer than a hundred lines of Firedrake/UFL, which we reproduce in abridged form; the examples can be reproduced with the freely available Firedrake and Fireshape packages.

2. Nonlinear Naghdi shell model

We adopt the five-parameter nonlinear Naghdi shell model: a shear-deformable, C^0 alternative to the Kirchhoff–Love shells common in isogeometric shell optimisation, well suited to a faceted triangulation and standard Lagrange elements. We begin by summarising its kinematics, its strain and stress measures, and the resulting variational problem. For a detailed treatment of the model, we refer the interested reader to [17, 18, 4]. The reference configuration of the shell is the swept surface

$$\mathbf{p}_0(x_\alpha, x_3) = \boldsymbol{\varphi}_0(x_\alpha) + x_3 \mathbf{n}_0(x_\alpha), \quad x_\alpha \in \omega \subset \mathbb{R}^2, \quad x_3 \in [-t/2, t/2], \quad (1)$$

where $\boldsymbol{\varphi}_0$ maps the parametric domain ω to the mid-surface and $\mathbf{n}_0$ is the unit normal. The deformed configuration is

$$\mathbf{p}(x_\alpha, x_3) = \boldsymbol{\varphi}(x_\alpha) + x_3 \mathbf{d}(x_\alpha), \quad \boldsymbol{\varphi} = \boldsymbol{\varphi}_0 + \mathbf{u}, \quad (2)$$

with $\mathbf{u}$ the mid-surface displacement and $\mathbf{d}$ the (unit, inextensible) director. Parameterising $\mathbf{d}$ by two angles $\boldsymbol{\beta} = (\beta_1, \beta_2) \in [-\frac{\pi}{2}, \frac{\pi}{2}] \times (-\pi, \pi]$ gives the five-parameter shell model

$$\mathbf{d}(\boldsymbol{\beta}) = (\sin \beta_2 \cos \beta_1, -\sin \beta_1, \cos \beta_2 \cos \beta_1). \quad (3)$$

The kinematics of the model is illustrated in Figure 1.

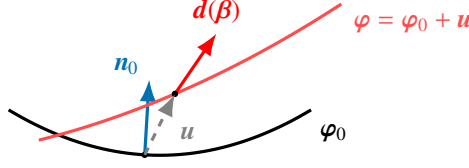


Figure 1: Kinematics of the nonlinear Naghdi shell: reference mid-surface φ_0 with unit normal $\mathbf{n}_0$, deformed mid-surface φ with director $\mathbf{d}(\beta)$ from (3), and the displacement $\mathbf{u}$.

These kinematics fix the strain and stress measures. With $\nabla\varphi = [\varphi_{,1} \ \varphi_{,2}]$ and reference metric $\mathbf{a}_0 = \nabla\varphi_0^\top \nabla\varphi_0$, the Lagrangian membrane, bending and shear strains are

$$\mathbf{e}(\mathbf{u}) = \frac{1}{2}(\nabla\varphi^\top \nabla\varphi - \mathbf{a}_0), \quad (4)$$

$$\mathbf{k}(\mathbf{u}, \beta) = -\frac{1}{2}(\nabla\varphi^\top \nabla\mathbf{d} + \nabla\mathbf{d}^\top \nabla\varphi) - \mathbf{b}_0, \quad (5)$$

$$\boldsymbol{\gamma}(\mathbf{u}, \beta) = \nabla\varphi^\top \mathbf{d} - \nabla\varphi_0^\top \mathbf{d}_0, \quad (6)$$

where $\mathbf{b}_0$ is the reference curvature. For a homogeneous St. Venant–Kirchhoff material the plane-stress elasticity tensor reads

$$A^{\alpha\beta\rho\sigma} = \frac{2\lambda\mu}{\lambda + 2\mu} a_0^{\alpha\beta} a_0^{\rho\sigma} + \mu(a_0^{\alpha\rho} a_0^{\beta\sigma} + a_0^{\alpha\sigma} a_0^{\beta\rho}), \quad (7)$$

giving stress resultants $\mathbf{N} = t\mathbf{A} : \mathbf{e}$, $\mathbf{M} = \frac{t^3}{12}\mathbf{A} : \mathbf{k}$ and $\mathbf{T} = t\mu\mathbf{a}_0^{-1}\boldsymbol{\gamma}$, and energy densities $\psi_m = \frac{1}{2}\mathbf{N} : \mathbf{e}$, $\psi_b = \frac{1}{2}\mathbf{M} : \mathbf{k}$, $\psi_s = \frac{1}{2}\mathbf{T} : \boldsymbol{\gamma}$.

Equilibrium then follows variationally, as the stationarity of the total potential

$$\Pi(\mathbf{u}, \beta) = \int_{\omega} (\psi_m + \psi_b + \psi_s) \, dx - W_{\text{ext}}, \quad (8)$$

where W_{ext} is the external load work. In this work, we consider two load models. For the semi-cylinder, the central point load on the free edge is regularised as a narrow Gaussian line load along that edge, $W_{\text{ext}} = \int_{\Gamma_{\text{free}}} P g(x) u_z \, ds$, where g is a unit-mass Gaussian of width $\sigma = \rho\pi/80$ in the circumferential coordinate $x = \rho \sin \theta$, centred at the crown $\theta = 0$, which avoids the pointwise singularity induced by point loads but converges to the point load as its width vanishes. For the bracket, a total force F is applied in the y -direction, distributed uniformly over the loaded edge, that is, $W_{\text{ext}} = \int_{\Gamma_{\text{load}}} (F/|\Gamma_{\text{load}}|) u_y \, ds$.

3. Finite element implementation

The distinguishing feature of the present implementation is that it operates on a *discrete* (faceted) triangulation $\mathcal{T}_h$ of the mid-surface, rather than on an exact analytic chart φ_0 ; every reference-geometry quantity is therefore recovered from the mesh. On each flat triangle, the affine geometric transformation that maps the finite element reference triangle onto the physical triangle has a constant Jacobian matrix J_0 . Specifically, J_0 is a 3×2 matrix whose two columns are the in-plane edge tangents. In this case, the reference metric (first fundamental form) is the constant $\mathbf{a}_0 = J_0^\top J_0$. The director is built from the surface normal. Each triangle carries a single constant unit normal, its *cell normal* $\mathbf{v}$ (UFL's `CellNormal`). Since these per-facet normals are discontinuous across shared edges, we recover a continuous normal field by L^2 -projecting $\mathbf{v}$ onto the continuous piecewise-linear space $[\text{CG}_1]^3$, which averages the adjacent facet normals at each vertex, and renormalising,

$$\mathbf{n}_0 = \frac{\mathcal{P}_h \mathbf{v}}{|\mathcal{P}_h \mathbf{v}|}, \quad \boldsymbol{\beta}_0 = (\text{atan2}(-n_{0,2}, \sqrt{n_{0,1}^2 + n_{0,3}^2}), \text{atan2}(n_{0,1}, n_{0,3})), \quad (9)$$

where $\mathcal{P}_h$ denotes the L^2 -projection onto $[\text{CG}_1]^3$ and $\boldsymbol{\beta}_0 = (\beta_{0,1}, \beta_{0,2})$ are the two director angles read off the recovered normal $\mathbf{n}_0$. The reference director then follows from these angles, $\mathbf{d}_0 = \mathbf{d}(\boldsymbol{\beta}_0)$ through (3), and the reference curvature (second fundamental form) is recovered consistently from its facet gradient,

$$\mathbf{b}_0 = -\frac{1}{2}(J_0^\top \nabla \mathbf{d}_0 + \nabla \mathbf{d}_0^\top J_0), \quad (10)$$

so that the metric $\mathbf{a}_0$, the curvature $\mathbf{b}_0$ and the director $\mathbf{d}_0$ are all obtained from the mesh alone. This recovered-geometry route is what makes the model compatible with shape optimisation: every design update produces a new mesh with no closed-form parameterisation, yet these reference quantities remain well defined. A consistent global orientation of the recovered normals is fixed once from a reference vector field, so that the director sign is coherent across the triangulation; the resulting geometric error is consistent and, for this recovered-geometry formulation, vanishes under mesh refinement [4] (Figure 8).

On this recovered geometry, displacements are discretised with bubble-enriched quadratics $\mathbf{u}_h \in [\text{CG}_2 \oplus B_3]^3$ and rotations with quadratics $\beta_h \in [\text{CG}_2]^2$. Membrane and shear locking are alleviated with a partial selective reduced integration (PSRI) scheme: the membrane and shear energies are split into full- and reduced-quadrature parts weighted by a cell factor α ,

$$\Pi_h = \Pi_b + \alpha(\Pi_m + \Pi_s)|_{\text{full}} + (1 - \alpha)(\Pi_m + \Pi_s)|_{\text{red}}, \quad \alpha = t^2/h^2, \quad (11)$$

with full quadrature of degree 4, reduced quadrature of degree 2, and h the cell diameter. The choice $\alpha \simeq t^2/h^2$ follows [4, 19] and balances the membrane/shear terms against the bending energy.

The resulting nonlinear residual is traced by load continuation: Newton's method (PETSc SNES, `newtonls` with an ℓ^2 line search and a MUMPS LU factorisation) under incremental loading $P \in \{0, \dots, P_{\max}\}$, using each converged state as the initial guess for the next increment.

All of this is expressed compactly in Firedrake. Listing 1 shows the core of the forward model. The residual and its Jacobian are obtained from the energy by automatic differentiation (`derivative`); no hand-coded tangent is required.

```
# Five-parameter director from two angles
def director(beta):
    return as_vector([sin(beta[1])*cos(beta[0]),
                     -sin(beta[0]),
                     cos(beta[1])*cos(beta[0])])

# Reference geometry recovered from the faceted mesh
J0 = Jacobian(mesh); a0 = J0.T*J0; a0_contra = inv(a0)
n0 = project(CellNormal(mesh), V_normal) # recovered normal
beta0 = interpolate(as_vector([atan2(-n0[1], sqrt(n0[0]**2 + n0[2]**2)),
                               atan2(n0[0], n0[2])]), V_beta) # initial director angles
d0 = director(beta0)

# Strain measures (membrane e, bending k, shear gamma)
F = dot(grad(u_), J0) + J0
d = director(beta_ + beta0)
e = 0.5*(F.T*F - a0)
k = -0.5*(F.T*dot(grad(d), J0) + dot(grad(d), J0).T*F) - b0
gamma = F.T*d - J0.T*d0

# PSRI: split membrane/shear into full (deg 4) and reduced (deg 2) parts
alpha = project(t**2/h**2, FunctionSpace(mesh, "DG", 0))
Pi = (psi_b*dx_full
      + alpha*(psi_m + psi_s)*dx_full
      + (1.0 - alpha)*(psi_m + psi_s)*dx_red)

R = derivative(Pi, q_, q_t) + W_ext # residual (auto-differentiated)
J = derivative(R, q_, q) # consistent tangent
solve(R == 0, q_, bcs=bcs, J=J, solver_parameters={...}) # solver parameters abridged
```

Listing 1: Core of the nonlinear Naghdi forward model in Firedrake (abridged).

4. Shape optimisation implementation

The shape-optimisation problem is PDE-constrained: the optimisation variable is the domain itself, and every objective evaluation requires a full nonlinear shell solve. Following [5, 20], this domain is represented as the image

of a bi-Lipschitz transformation $\mathbf{T} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ applied to an initial domain Ω , so that the set of admissible domains is

$$\mathcal{Q} = \{ \mathbf{T}(\Omega) : \mathbf{T} \text{ bi-Lipschitz} \}. \quad (12)$$

We minimise the reduced objective $\hat{J}(\mathbf{T}) := J(\mathbf{T}, \mathbf{u}(\mathbf{T}))$ subject to the nonlinear Naghdi equations of Section 2. Two objectives are used, matching each benchmark to its own measure of stiffness. For the semi-cylinder showcase we minimise the *squared-displacement functional* $J_{\text{sc}} := \int_{\omega} |\mathbf{u}|^2 dx$, a direct measure of the kinematic response under the load; for the engineering bracket case we reproduce the COMSOL benchmark, which minimises the total elastic strain energy (compliance) $J_{\text{br}} := \int_{\omega} (\psi_m + \psi_b + \psi_s) dx$, scaled by its initial value. Additionally, to reflect the spirit of the applications, we introduce the (differentiable) penalty term

$$P(\mathbf{T}) := \frac{1}{2} \int_{\omega} [(|\mathbf{T} - \mathbf{id}| - \delta_{\max})_+]^2 dx, \quad (s)_+ := \max(s, 0), \quad (13)$$

which penalises pointwise shape displacements in excess of a prescribed budget $\delta_{\max}$ and so limits the admissible shape change. In the implementation the norm and the positive part are smoothed with a small parameter ε (e.g. $|\cdot| \rightarrow \sqrt{|\cdot|^2 + \varepsilon^2}$), and the constant offset is subtracted, so that P is everywhere differentiable and $P(\mathbf{id}) = 0$. Hence, we either minimise

$$J = J_{\text{sc}} + \gamma P \quad \text{or} \quad J = J_{\text{br}} + \gamma P, \quad (14)$$

where $\gamma > 0$ is a penalisation parameter. The use of different objectives for the two test cases simply reflects the two applications; the same adjoint pipeline handles either objective.

The optimiser updates are driven by the shape derivative $d\hat{J}$ of the reduced objective $\hat{J}$. Shape derivatives are computed by algorithmic differentiation (UFL and pyadjoint, [3, 21]): the forward load-continuation solve is taped and the adjoint is replayed to assemble $d\hat{J}$. Because the shell is a two-dimensional manifold embedded in $\mathbb{R}^3$, the feasibility of a trial shape is measured on each cell by the surface Jacobian of the moved element map (Section 3), $\sqrt{\det(J_0^T J_0)} = |\mathbf{j}_{0,1} \times \mathbf{j}_{0,2}|$ — equivalently $\sqrt{\det \mathbf{a}_0}$, the elementwise area scale factor, with $\mathbf{j}_{0,1}$ and $\mathbf{j}_{0,2}$ the two tangent columns of J_0 evaluated on the moved mesh — which must remain strictly positive; the mesh quality of the optimised designs is verified in Section 5.4. Only optimiser-accepted shapes are taped to keep the tape consistent with the trust-region state, and trial shapes that are subsequently rejected are evaluated without annotation. This keeps the adjoint memory footprint bounded by a single continuation history regardless of the number of trial steps.

To discretise the set of admissible domains $\mathcal{Q}$, we consider transformations of the form $\mathbf{T} = \mathbf{id} + \mathbf{d}$, where $\mathbf{d}$ is a boundary-preserving vector field that displaces the mesh interior. It remains to fix the *discrete control space* $\mathcal{Q}$ in which the design displacement $\mathbf{d}$ lives and the *inner product* $a(\cdot, \cdot)$ used to compute the shape gradient $\nabla \hat{J}$, which informs how the optimiser updates $\mathbf{d}$ and is the solution to

$$a(\nabla \hat{J}, \mathbf{W}) = d\hat{J}(\mathbf{W}) \quad \text{for all } \mathbf{W} \in \mathcal{Q}. \quad (15)$$

Because the optima of interest are smooth, large-scale shape changes, we select a lower-dimensional *multilevel* (geometric-multigrid) control equipped with a smoothing inner product, so the optimiser commits to coherent folds rather than dithering among small-amplitude, high-frequency minima [5, 22]. Let a mesh hierarchy be obtained by uniform refinement of a coarse triangulation $\mathcal{T}_0$ of the mid-surface, $\mathcal{T}_0 \rightarrow \cdots \rightarrow \mathcal{T}_L$, with nested vector coordinate spaces $V_0 \subset \cdots \subset V_L$. We take the *control* on the coarsest level, $\mathcal{Q} = V_0$, and solve the shell state on the *finest* level $\mathcal{T}_L$. Writing $P : V_0 \rightarrow V_L$ for the *prolongation* (the composition of inter-grid transfers), the deformation of the finest mesh is given by

$$\mathbf{T} = \mathbf{id} + P \mathbf{q}, \quad \mathbf{q} \in V_0. \quad (16)$$

The shape derivative of $\hat{J}$ is first evaluated along mesh deformations of the fine mesh using the adjoint method. Then, applying the transpose of the prolongation P (the standard multigrid restriction), we compute its *restriction* to the coarse control space. To compute the shape gradient $\nabla \hat{J} \in V_0$ we take the inner product a to be a *smoothing* (Helmholtz) metric,

$$a(g, v) = \int_{\Omega} (g \cdot v + \ell^2 \nabla g : \nabla v) dx, \quad (17)$$

which acts as a Helmholtz filter of feature size ℓ , closely related to the Helmholtz/PDE filters and vertex-morphing regularisation used in node-based structural shape optimisation [23, 24]: a larger length scale ℓ penalises short-wavelength control variation more strongly and so sets the wavelength of the folds. Because $\dim V_0 \ll \dim V_L$ the design space automatically excludes high-frequency designs, while the fine state mesh enables accurate computation of the energy and its gradient using the faceted forward model described in Section 3. In Fireshape this approach is implemented via the `FeMultiGridControlSpace` with a Helmholtz `UflInnerProduct` (Listing 2).

```
class ShellShapeObjective(PDEconstrainedObjective):
    def objective_value(self):
        self.update_geometry_coefficients() # recompute n0, beta0, alpha
        self.q_.assign(0.0)
        for P in self.load_values[1:]: # load continuation
            self.P_load.assign(float(P)); self.solver.solve()
        u_h, _ = self.q_.subfunctions
        return assemble(self.psi_total*dx) # compliance / energy objective

class HelmholtzInnerProduct(UflInnerProduct): # smoothing Riesz map (M + l^2 K)
    def get_weak_form(self, V):
        u, v = TrialFunction(V), TestFunction(V)
        return (self.l**2*inner(grad(u), grad(v)) + inner(u, v))*dx

# Multilevel control on a mesh hierarchy: coarse control, fine state
mh = MeshHierarchy(base_mesh, refine)
Q = FeMultiGridControlSpace(mh, coarse_control=True)
inner = HelmholtzInnerProduct(Q, length_scale=..., fixed_bids=FIXED_TAGS,
                              extra_bcs=symmetry_bcs, direct_solve=True)
q = ControlVector(Q, inner)
J = ShellShapeObjective(Q, ...) + ShapeDisplacementPenalty(Q, limit=..., scale=...)

# Trust-region BFGS via ROL
solver = ROL.OptimizationSolver(ROL.OptimizationProblem(J, q), params)
solver.solve()
```

Listing 2: Coupling the shell to Fireshape and ROL with a multilevel control and a Helmholtz smoothing metric (abridged).

The resulting optimisation problem is solved with the optimisation library ROL [25], using a trust-region method, a limited-memory BFGS Hessian approximation, and a truncated-CG subproblem solver. Figure 2 shows the overall optimisation loop.

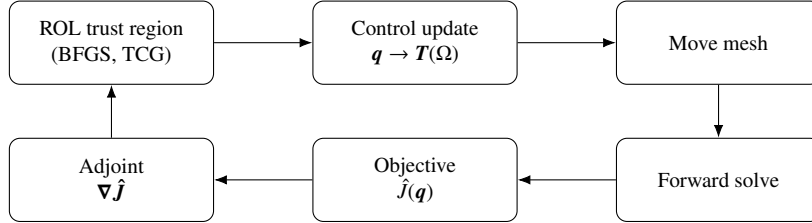


Figure 2: PDE-constrained shape-optimisation loop coupling ROL, Fireshape (control space and mesh motion) and the Firedrake nonlinear Naghdi forward and adjoint solves.

We close with the numerical settings common to the two examples. In both, the state fields are discretised with bubble-enriched quadratic displacements $[CG_2 \oplus B_3]^3$ and quadratic rotations $[CG_2]^2$ as in Section 3, and the discretised control variable is a piecewise-linear coordinate field on the coarsest mesh in the hierarchy. The semi-cylinder is optimised on the half model ($\theta \in [0, \pi/2]$), exploiting its mirror symmetry (Section 5.3): a 1024-triangle control mesh, refined uniformly once to the 4096-triangle state mesh (54,213 state degrees of freedom), with the load raised to 2 kN in 40 continuation steps and Helmholtz length scale $\ell = 1.0$ m. The bracket is optimised on its half model, exploiting the mid-plane mirror symmetry (Section 5.2), with a 6080-triangle control mesh and a 24,320-triangle state mesh (319,605 degrees of freedom), the load raised to 10 kN (applied as 5 kN on the half) in 5 continuation steps and $\ell = 0.6$ m. Each continuation step is solved by Newton’s method (PETSc `newtonls`, ℓ^2 line search, MUMPS LU)

to relative and absolute residual tolerances 10^{-8} and 10^{-9} (at most 100 iterations). The reduced problem is solved by the ROL trust-region method with a limited-memory BFGS Hessian and a truncated-CG subproblem solver (initial trust-region radius 0.01, radius growing rate 2, step-acceptance ratio 10^{-3}), stopped at a gradient tolerance of 10^{-3} , a step tolerance of 10^{-7} , or 50 iterations. The soft shape-displacement penalty (13) uses weight $\gamma = 10^5$ and smoothing $\varepsilon = 5 \times 10^{-3}$ m, with the per-case budget $\delta_{\max}$ listed in Tables 1 and 2.

5. Results

5.1. Forward-model validation against Abaqus

We validate the forward solver against the Sze semi-cylindrical shell benchmark, using the tabulated Abaqus S4R reference values of [15]. The model is a semi-cylindrical (half-cylinder) shell whose mid-surface $\{(\rho \sin \theta, y, \rho \cos \theta) : \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}], y \in [0, L]\}$ has radius $\rho = 1.016$ m, axial length $L = 3.048$ m and uniform thickness $t = 30$ mm, and whose material has Young’s modulus $E = 20.685$ MPa and Poisson ratio $\nu = 0.3$ (Figure 4a). Of the two curved edges, the one at $y = 0$ is fully clamped ($\mathbf{u} = \mathbf{0}, \boldsymbol{\beta} = \mathbf{0}$) while the opposite edge at $y = L$ is left free; the two straight lateral edges $\theta = \pm \frac{\pi}{2}$ carry the mirror-symmetry conditions $u_z = 0$ and $\beta_2 = 0$ that let this half model represent the full cylinder. A point load P , directed downward and raised by load continuation to a maximum of 2 kN, is applied at the crown $\theta = 0$ — the centre of the free curved edge. These geometric, material and loading parameters are summarised in Table 1.

The computed load–deflection curve (Figure 3) reproduces the reference response across the full load range, including the stiffening beyond ≈ 750 N. The agreement is quantitative: the root-mean-square error against the reference points is $8.601\text{e-}03$ m (0.5 % of the peak deflection), with a maximum error of $1.510\text{e-}02$ m.

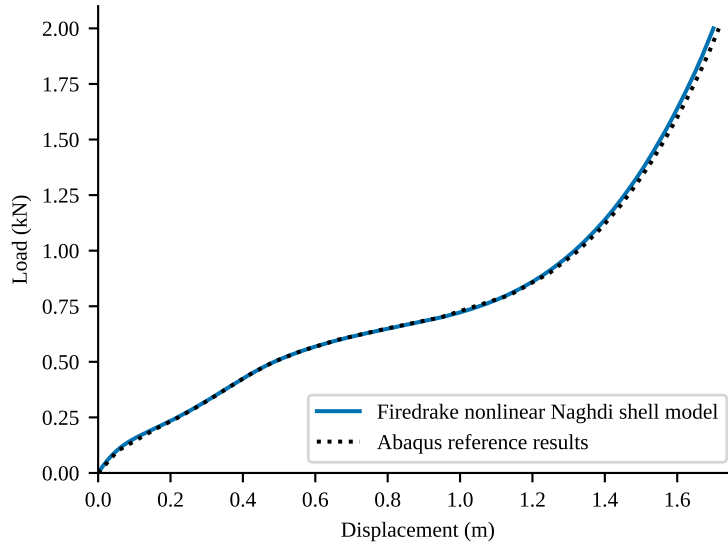


Figure 3: Clamped semi-cylindrical shell under point load: load–deflection of the downward displacement at the load point, Firedrake nonlinear Naghdi model vs. the Sze/Abaqus S4R reference [15].

5.2. Shape-optimisation validation against COMSOL: sheet-metal bracket

We first validate the shape-optimisation capability against the COMSOL “Shape Optimization of a Shell” benchmark [16]. The problem data (Figure 6a) is an L-shaped steel shell ($E = 200$ GPa, $\nu = 0.3$, $t = 0.01$ m) with a 0.3 m fillet, loaded by a total force of 10 kN in the y -direction on the free edge and clamped on the opposite edge, optimised for maximum stiffness (minimum total elastic strain energy) under geometric nonlinearity. The design boundary is constrained to follow the benchmark: under the shape update the clamped edge may slide only within its own plane (a roller, $u_y^s = 0$) while the remaining external edges are held fixed ($\mathbf{u}^s = \mathbf{0}$), and the admissible shape change is limited to 0.05 m.

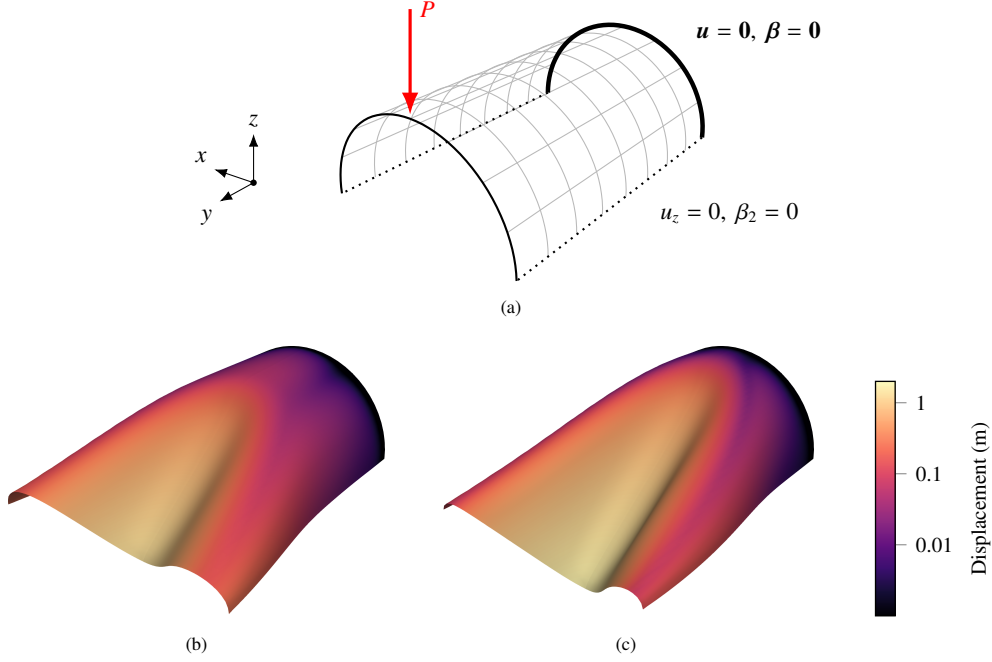


Figure 4: Forward model and response of the reference semi-cylinder. (a) The semi-cylindrical shell benchmark used as the running example: a half-cylinder mid-surface $\{(\rho \sin \theta, y, \rho \cos \theta) : \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}], y \in [0, L]\}$ of radius ρ , axial length L and thickness t , faceted into the triangulation $\mathcal{T}_h$, fully clamped along the $y = 0$ end ($\mathbf{u} = \mathbf{0}, \boldsymbol{\beta} = \mathbf{0}$) and free along the $y = L$ end, with mirror-symmetry conditions $u_z = 0$ and $\beta_2 = 0$ on the two straight lateral edges ($\theta = \pm \frac{\pi}{2}$) and a downward point load P at the crown ($\theta = 0$) of the free edge. (b, c) Forward response computed on the same mesh as the shape optimisation: the deformed configuration (mesh warped by the displacement) coloured by displacement magnitude (magma, shared scale and camera) at (b) 1 kN and (c) 2 kN, viewed from the loaded free end; the load-point deflection grows strongly and nonlinearly, reaching 1.41 m at 1 kN and 1.81 m at 2 kN. Benchmark parameters are listed in Table 1.

The optimisation uses the multilevel (coarse-grid) control with the Helmholtz smoothing metric of Section 4 at length scale $\ell = 0.6$ m and a relaxed displacement-penalty weight, solved on the half model ($z \in [0, \frac{1}{2}H]$) with a symmetry-plane condition and mirrored to the full bracket. The optimiser moves material away from the mid-plane, forming a stiffening corrugation (Figures 5–6) that recruits bending stiffness from the initially flat regions. It reduces the elastic strain energy by 87 % while travelling at most 0.032 m — within the 0.05 m budget — and increasing the surface area by +8.3 %, essentially matching COMSOL’s 89 % reduction and +9 % area change (Table 2). The automated nonlinear-Naghdi framework thus attains COMSOL-magnitude stiffening, at the same area change and within the same budget, on the same benchmark.

5.3. Shape optimisation of a curved structure: the semi-cylinder

Having validated the shape optimisation against COMSOL on the bracket, we apply the framework to the curved semi-cylinder, whose forward response was validated against Abaqus in Section 5.1. We minimise the (squared) displacement under the crown point load with the same ingredients used for the bracket: a multilevel (coarse-grid) control and a Helmholtz smoothing metric (Section 4) at length scale $\ell = 1.0$ m on the validated faceted forward model. As with the bracket, the problem is symmetric — here about the plane $x = 0$ through the crown and the load — so we optimise the half model ($\theta \in [0, \pi/2]$) with a mirror-plane condition and reflect the result; a weak penalty on the in-plane slope of the control at the mirror plane keeps the reflected design C^1 -continuous across the crown. The control lives on the coarse arc (resolved at the validated facet density) and the state on its uniform refinement.

The optimiser redistributes the mid-surface into a smooth stiffening crease along the crown near the loaded free end (Figure 7), keeping the shape displacement within the imposed 0.2 m limit (maximum 0.177 m). Under the 2 kN crown load this reduces the squared-displacement functional by 91.0 % (Figure 5b) relative to the reference cylinder — equivalently, the shell’s average deflection falls by 78 % (from 0.36 m to 0.08 m) — at a modest +4.1 % change of

Table 1: Clamped semi-cylindrical shell benchmark: geometry, material, conditions, validation of the load-point deflection against the Sze/Abaqus reference [15] (errors as a percentage of the peak deflection), and the Fireshape shape-optimisation results.

Clamped semi-cylindrical shell	
<i>Geometry</i>	
Radius ρ	1.016 m
Axial length L	3.048 m
Thickness t	30 mm
<i>Material</i>	
Young’s modulus E	20.685 MPa
Poisson ratio ν	0.3
<i>Conditions</i>	
Applied point load P	2 kN
Shape-displacement budget	200 mm
<i>Load-point deflection error (vs. Abaqus)</i>	
Displacement RMSE	0.5 %
Maximum displacement error	0.9 %
<i>Shape optimisation</i>	
Displacement-functional reduction	91.0 %
Average-deflection reduction	78 %
Maximum shape displacement	177 mm
Added surface area	4.1 %

surface area, demonstrating that the same adjoint shape-derivative pipeline delivers a large, smooth stiffening gain on a curved structure.

5.4. Numerical verification and robustness

We close the results with verification and robustness checks of the discretisation, the adjoint gradient and the optimisation.

Mesh convergence. Refining the faceted semi-cylinder mesh across thirteen resolutions from 128 to 12800 cells drives the load-point deflection at 2 kN monotonically towards the reference (from 1.61 m at the coarsest mesh to 1.710 m, against the Abaqus 1.715 m), the RMSE against the S4R data falling from 5.0 % to a 0.30 % floor — the residual model difference rather than a discretisation error (Figure 8); the 800-cell validation mesh is already within 0.5 %. On the optimisation side, evaluating the loaded-shape objective $\int_{\omega} |\mathbf{u}|^2 dx$ on the 4096-cell state mesh and on one further uniform refinement (16384 cells) changes it by only 0.068 %, so the reported displacement-functional reduction is mesh-independent.

Adjoint shape derivative. A Taylor test of the reduced objective — perturbing the mesh deformation and comparing the objective against its adjoint-derived first-order model — shows second-order convergence (observed rates 1.7 to 1.9 over four step halvings, the mild drop at the finest step reflecting the nonlinear-solver tolerance), confirming the shape derivative assembled by algorithmic differentiation through the load-continuation solve.

Mesh quality and load regularisation. Complementing the surface-Jacobian feasibility measure of Section 4, we verify the optimised meshes directly: both optima are free of inverted or degenerate cells, with minimum normalised triangle quality 0.71 (semi-cylinder) and 0.73 (bracket) and minimum angle $\approx 31^\circ$, down only modestly from $\approx 44^\circ$ initially. The point-load regularisation is likewise benign: halving the Gaussian half-width (from $\rho\pi/80$ to $\rho\pi/160$) changes the 2 kN deflection by 0.21 %, and a fourfold range of widths spans under 0.4 %, confirming that the regularised load approximates the point-load limit.

Role of the smoothing length scale. The smoothing metric governs the fold wavelength and, in this large-deflection regime, the feasibility of the optimisation trajectory. At $\ell = 1.0$ m the 2 kN optimisation converges to the 91.0 %

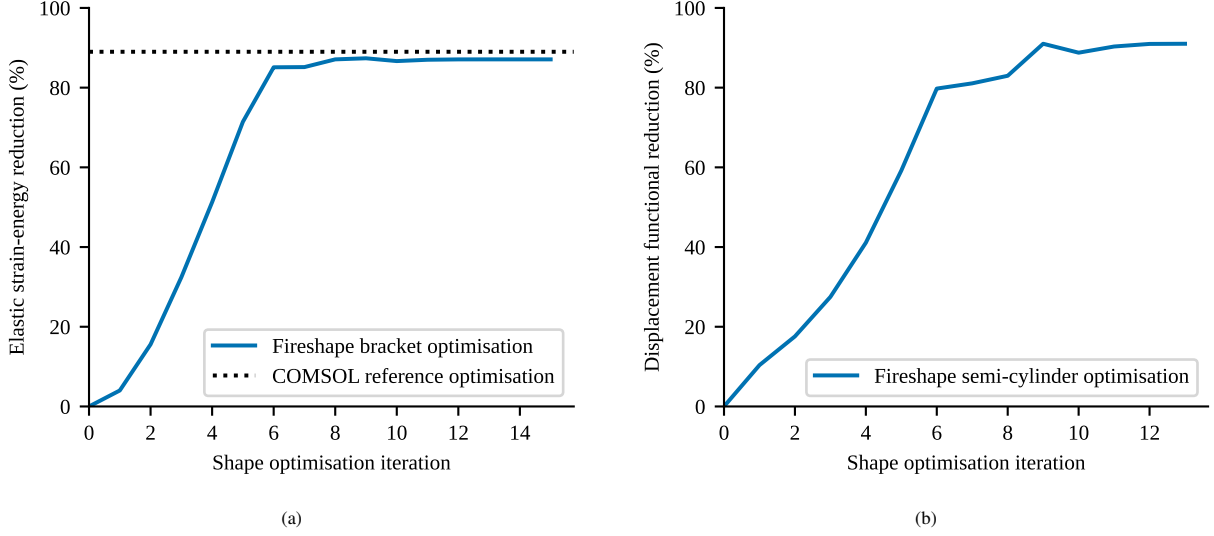


Figure 5: Shape-optimisation convergence — objective reduction versus iteration. (a) Sheet-metal bracket: elastic strain-energy reduction; the symmetric multilevel control with the smoothing metric reaches 87 %, approaching the COMSOL benchmark (dotted). (b) Semi-cylinder: squared-displacement functional reduction under the 2 kN crown load, reaching 91.0 % within the 0.2 m shape budget.

reduction of Section 5.3; re-running at $\ell = 0.6$ m and $\ell = 1.4$ m the forward continuation instead broke down at intermediate trial shapes before convergence, so $\ell = 1.0$ m is retained as the robust operating point at this load.

Computational cost. Table 3 lists the cost of a forward solve and its adjoint on the production discretisation. The strongly nonlinear 2 kN response is traced by load continuation — five increments are the fewest that converge — which dominates at 92 s on six MPI ranks, whereas the adjoint shape derivative, a linear replay of the taped continuation, costs only 3.5 s, under 4 % of the forward. The optimisation itself uses more increments to keep the rougher intermediate shapes robust, but each design iteration remains dominated by these forward evaluations.

6. Discussion

The polygonal geometry discretisation is sufficiently accurate and it enables shape optimisation. The forward model attains an RMSE of $8.601\text{e-}03$ m (0.5 % of the peak deflection) against the Abaqus reference, despite recovering the metric, curvature and director numerically from a faceted triangulation rather than from an analytic chart. The faceting introduces a geometric error that is consistent and, for this formulation, decreases under mesh refinement [4] (Figure 8); the benchmark test shows that the differences among simulated responses are already negligible at engineering resolution. Crucially, this recovered-geometry route is what makes shape optimisation tractable: each design update yields a new mesh with no closed-form parameterisation, yet all shell quantities remain well defined and differentiable. The PSRI split $\alpha = t^2/h^2$ is recomputed per cell for every shape, so locking control remains consistent as the mesh deforms.

The cost of the shape gradient follows directly from this nonlinear forward model. Because the forward problem is solved by load continuation, the taped program contains every increment, and the adjoint replays them in reverse; the marginal cost of a gradient is therefore of the order of one additional linear solve per load step. Taping only optimiser-accepted shapes (Section 4) avoids accumulating rejected trial states and keeps the memory footprint bounded.

Beyond this cost, the control space, smoothing and symmetry govern the attainable stiffness gain. The bracket case (Section 5.2) shows that, for the *same* physics and constraints, the achieved reduction depends strongly on how the design is parameterised, smoothed and symmetrised. The nodal control space, paired with a domain-tied elasticity metric, favours smooth low-amplitude shapes and converges to a 68 % local minimum (at +5.2 % area) that does *not* improve with mesh refinement. Restricting the search to the smooth, large-scale modes of a multilevel control space (Section 4), equipped with a Helmholtz smoothing metric whose length scale sets the fold wavelength, is

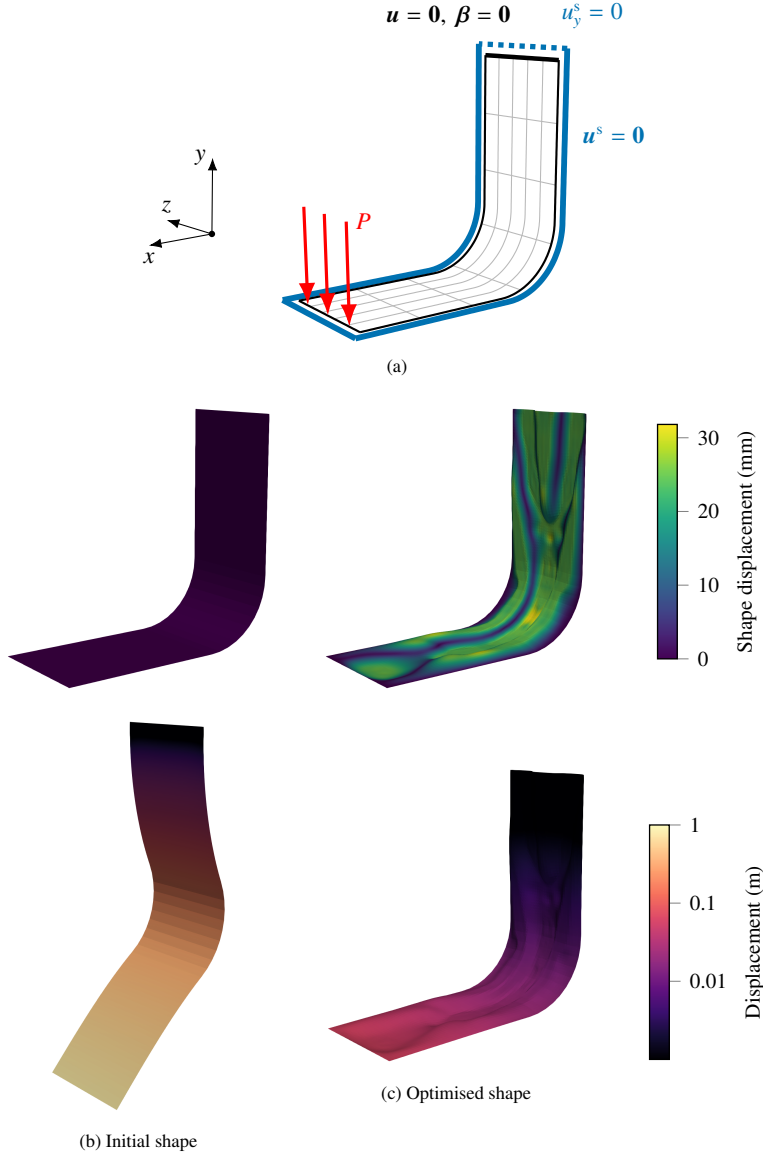


Figure 6: Bracket shape optimisation with the symmetric multilevel control space (half model mirrored about the mid-plane). (a) The shape-optimisation set-up on the full bracket — an L-shaped shell (vertical and horizontal legs joined by a fillet, extruded in z) with the forward clamped edge ($u = 0, \beta = 0$) and the 10 kN load P in $-y$ on the free edge (black/red), together with the fixed-shape boundary condition (blue): $u^s = 0$ (all components, bold) on the loaded edge and the two extrusion rims, and the roller $u_y^s = 0$ (only the y component, dotted) on the clamped edge, so the boundary is held while the interior is reshaped. Middle row: the reference mid-surface coloured by shape-displacement magnitude (viridis) for the initial (left) and optimised (right) design (87 % strain-energy reduction). Bottom row: the corresponding deformed configuration under the 10 kN load (mesh warped by the displacement) coloured by displacement magnitude (magma) for (b) the initial and (c) the optimised design. The optimised off-mid-plane corrugation recruits bending stiffness, so the loaded design (c) deflects far less than the initial one (b) — reproducing the mechanism of the COMSOL benchmark [16].

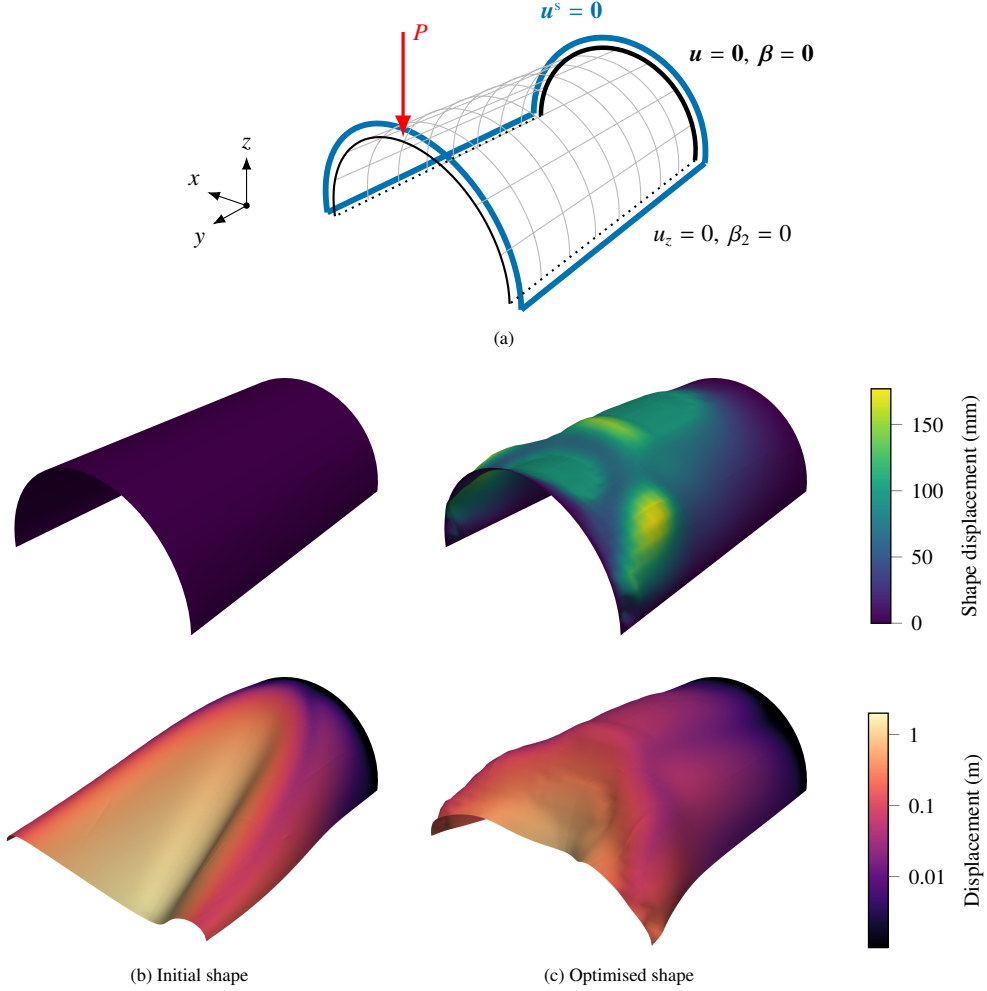


Figure 7: Semi-cylinder shape optimisation with the multilevel control and the Helmholtz smoothing metric (half model mirrored about the $x = 0$ symmetry plane), viewed from the loaded free end. (a) The shape-optimisation set-up on the full semi-cylinder — the forward clamped and lateral-symmetry conditions and the crown load ($\mathbf{u} = \mathbf{0}, \boldsymbol{\beta} = \mathbf{0}; u_z = 0, \beta_2 = 0; P$, in black/red) together with the fixed-shape boundary condition (blue outline): the shape displacement is held fixed, $\mathbf{u}^s = \mathbf{0}$, on every rim — the clamped end, the free loaded end and the two lateral edges — so the optimiser reshapes only the interior mid-surface while the boundary stays put. Middle row: the reference mid-surface coloured by the magnitude of the shape displacement (how far the optimiser moved each point) for the initial (left) and optimised (right) design (shared viridis scale). Bottom row: the corresponding deformed configuration at the maximum load of 2 kN — the mesh warped by the loaded displacement — coloured by the displacement magnitude for (b) the initial and (c) the optimised design (shared magma scale). The optimiser forms a smooth, symmetric stiffening crease along the crown near the loaded free end, reducing the squared-displacement functional by 91.0 %: the optimised design (c) deflects far less than the initial one (b).

Table 2: Sheet-metal bracket (COMSOL “Shape Optimization of a Shell” benchmark [16]): geometry, material, loading and the symmetry-enforced multilevel shape-optimisation result, with the COMSOL benchmark value in parentheses. All present results use the geometrically nonlinear Naghdi model with five load-continuation steps.

Sheet-metal bracket	
<i>Geometry</i>	
Footprint L	1.0 m
Height H	0.5 m
Fillet radius R	0.3 m
Thickness t	10 mm
<i>Material</i>	
Young’s modulus E	200 GPa
Poisson ratio ν	0.3
<i>Conditions</i>	
Applied force	10 kN
Shape-displacement budget	50 mm
<i>Shape optimisation (vs. COMSOL)</i>	
Strain-energy reduction	87 % (89 %)
Maximum shape displacement	32 mm (50 mm)
Added surface area	8.3 % (9 %)

Table 3: Cost of a forward solve and adjoint shape derivative for the semi-cylinder at 2 kN (4096-cell state mesh, 54,213 degrees of freedom, six MPI ranks); five load-continuation increments.

Computational cost	
Forward solve (5-step load continuation)	92 s
Adjoint shape derivative	3.5 s
Adjoint/forward cost ratio	3.8 %

what lets the optimiser commit to the deep off-mid-plane corrugation; a relaxed penalty — whose nominal weight otherwise imposes a below-limit drag scaling as $(\text{smoothing})^2 \times \text{weight}$ — frees its amplitude. A further, instructive point concerns *symmetry*. The problem is symmetric about its mid-plane, yet a full-model optimisation breaks that symmetry: the discrete mesh is not perfectly symmetric, and because the corrugation optimum is symmetry-unstable the optimiser amplifies the perturbation into a bumpy, over-travelling shape (maximum displacement 0.060 m, area +12.8 %). Enforcing the symmetry on a half model removes both pathologies at once and yields a clean, exactly symmetric design that reaches 87 % *within* the 0.05 m budget, at COMSOL’s area change. The small residual gap to COMSOL’s 89 % is due to constraint handling rather than mechanism: COMSOL searches *at* a hard displacement bound with a filtered free-shape, whereas we relax toward the bound with a soft penalty; a hard box constraint on the shape displacement is the natural way to recover the last two points and is identified as future work. We note that the smoothing metric is effective precisely because it acts on the lower-dimensional multilevel control; the same filter applied to the full nodal control was poorly conditioned and stalled early. The same recipe — a multilevel control, a Helmholtz smoothing metric and enforced symmetry — carries over unchanged to the curved semi-cylinder (Section 5.3), where it cuts the squared-displacement functional by 91.0 %; the stiffening mechanism is therefore not specific to the bracket but a general capability of the framework. There a length-scale sweep selects a longer $\ell = 1.0$ m for the smoothest large-scale fold, and because the multigrid hierarchy refines the curved reference coplanarly — adding deformation resolution but no geometric fidelity — the coarse control must already resolve the arc at the validated facet density.

This raises the question of why we adopt a multilevel control rather than a higher-order one. A natural alternative to a coarse control is one on *fewer but higher-order* elements, which curves each element interior and reduces faceting.

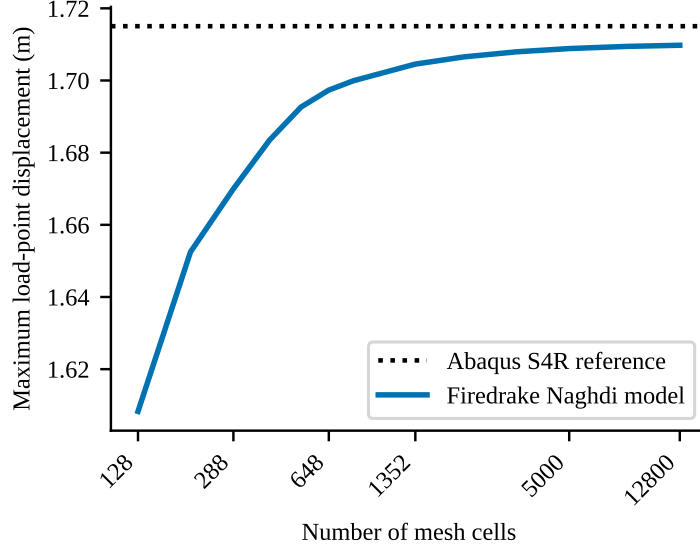


Figure 8: Forward-model mesh convergence: load-point deflection of the semi-cylinder at 2 kN versus the number of faceted mesh cells, converging to the Sze/Abaqus S4R reference. The 800-cell validation mesh is already within 0.5 %.

We avoid it for three reasons. First, and decisively, the validated forward model is affine: it recovers the director per facet and sets the PSRI split from a per-cell CellDiameter (Section 3). A higher-order control turns the moving mesh into a curved isoparametric element, on which these quantities are undefined or unsupported. Second, smoothness is more naturally imposed through the *metric* rather than through the polynomial degree: the Helmholtz Riesz map associated to Eq. (17) provides a single, continuously tunable length scale ℓ , decoupled from the number of design degrees of freedom and from the state resolution, whereas raising the element order entangles all three and — being only C^0 — would not even remove the inter-element slope discontinuities it targets. Third, the recovered-director shell is *validated as faceted* (Section 5.1), so residual geometric faceting of the control is not a modelling error; refining the coarse control by one level (the affine analogue of adding resolution) changes the bracket optimum by less than a quarter of a percent, confirming that faceting is not the limiting factor. A multilevel P_1 control with a smoothing metric thus retains the validated forward model, the standard multigrid transfer and its exact adjoint, and a clean separation of design dimension, smoothness and physics resolution.

Some limitations remain. The present study considers a single, conservative, quasi-static load case and minimises a stiffness objective without stability, stress or manufacturing constraints. For sheet-metal forming in particular, frictional self-contact and a path-dependent material response would be required to predict the formed shape faithfully; here the bracket is treated purely as an elastic-stiffness design problem. These restrictions are not intrinsic to the framework and define the natural next research steps.

7. Conclusions

We have presented an automated framework for the shape optimisation of geometrically nonlinear thin shells. Its distinguishing feature is a five-parameter Naghdi shell formulated on a discrete, faceted mesh with a numerically recovered director field, which removes the reliance on an analytic mid-surface parameterisation and so makes the geometry itself a natural design variable. Expressed in fewer than a hundred lines of Firedrake/UFL, the model yields its residual, tangent and adjoint automatically and couples directly to the Fireshape diffeomorphism control space and to ROL.

The framework was validated end to end: the forward solver reproduces the Sze/Abaqus semi-cylinder benchmark to 0.5 %, and on the COMSOL “Shape Optimization of a Shell” benchmark — an industrially relevant sheet-metal bracket — the optimisation develops the expected off-mid-plane corrugation and attains an 87 % reduction of

elastic strain energy *within* the benchmark’s displacement budget and at its area change, matching the benchmark’s magnitude. A multilevel control space with a smoothing metric proved essential: restricting the search to smooth, large-scale modes lets the optimiser reach the deep corrugation that a nodal control, biased toward small-amplitude, high-frequency local minima, misses. Applying the same validated recipe to the curved semi-cylinder — the shell used for the forward validation — cut its squared-displacement functional by 91.0 %, showing that the capability generalises beyond the bracket benchmark.

A second, instructive finding concerns *symmetry*. The bracket problem is symmetric about its mid-plane, but a full-model optimisation breaks that symmetry — seeded by the asymmetric mesh and amplified because the corrugation optimum is symmetry-unstable — into a bumpy, over-travelling shape. Enforcing the symmetry on a half model recovers a clean, exactly symmetric design that is stiffer per unit of shape displacement and stays within budget. The small residual gap to COMSOL’s 89 % is one of constraint handling: COMSOL searches at a hard displacement bound, whereas we relax toward it with a soft penalty.

Future work follows directly from these findings and the limitations above: a hard box constraint on the shape displacement (to search at the budget, as the benchmark does) to close the final gap; multiple and non-conservative load cases; stability- and stress-constrained objectives; frictional self-contact and path-dependent plasticity for genuine sheet-metal forming; and richer B-spline/free-form-deformation control spaces.

CRediT authorship contribution statement

Ado Farsi: Conceptualization, Methodology, Software, Validation, Investigation, Visualization, Writing — original draft. **Alberto Paganini:** Writing — review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Code and data availability

The nonlinear Naghdi shell forward model and the shape-optimisation examples presented in this paper are implemented entirely with open-source software and can be reproduced using the finite-element library Firedrake (<https://github.com/firedrakeproject/firedrake>) together with the shape-optimisation toolbox Fireshape (<https://github.com/fireshape/fireshape>). The example scripts are available from the corresponding author upon reasonable request.

Acknowledgements

Ado Farsi’s work on this project was supported by the Impact Development Fund (2025/26) of the University of Leicester.

References

- [1] F. Rathgeber, D. A. Ham, L. Mitchell, M. Lange, F. Luporini, A. T. T. McRae, G.-T. Bercea, G. R. Markall, P. H. J. Kelly, Firedrake: automating the finite element method by composing abstractions, *ACM Transactions on Mathematical Software* 43 (3) (2016) 1–27. doi:10.1145/2998441.
- [2] M. S. Alnæs, A. Logg, K. B. Ølgaard, M. E. Rognes, G. N. Wells, Unified Form Language: A domain-specific language for weak formulations of partial differential equations, *ACM Transactions on Mathematical Software* 40 (2) (2014) 1–37. doi:10.1145/2566630.
- [3] S. K. Mitusch, S. W. Funke, J. S. Dokken, dolfin-adjoint 2018.1: automated adjoints for FEniCS and Firedrake, *Journal of Open Source Software* 4 (38) (2019) 1292. doi:10.21105/joss.01292.
- [4] J. S. Hale, M. Brunetti, S. P. A. Bordas, C. Maurini, Simple and extensible plate and shell finite element models through automatic code generation tools, *Computers & Structures* 209 (2018) 163–181. doi:10.1016/j.compstruc.2018.08.001.
- [5] A. Paganini, F. Wechsung, Fireshape: a shape optimization toolbox for Firedrake, *Structural and Multidisciplinary Optimization* 63 (2021) 2553–2569. doi:10.1007/s00158-020-02813-y.

- [6] Y. Ding, Shape optimization of structures: a literature survey, *Computers & Structures* 24 (6) (1986) 985–1004.
- [7] K.-U. Bletzinger, E. Ramm, Form finding of shells by structural optimization, *Engineering with Computers* 9 (1) (1993) 27–35.
- [8] W. A. Wall, M. A. Frenzel, C. Cyron, Isogeometric structural shape optimization, *Computer Methods in Applied Mechanics and Engineering* 197 (33–40) (2008) 2976–2988.
- [9] S. Cho, S.-H. Ha, Isogeometric shape design optimization: exact geometry and enhanced sensitivity, *Structural and Multidisciplinary Optimization* 38 (1) (2009) 53–70.
- [10] J. Kiendl, R. Schmidt, R. Wüchner, K.-U. Bletzinger, Isogeometric shape optimization of shells using semi-analytical sensitivity analysis and sensitivity weighting, *Computer Methods in Applied Mechanics and Engineering* 274 (2014) 148–167.
- [11] T. Hirschler, R. Bouclier, A. Duval, T. Elguedj, J. Morlier, The embedded isogeometric Kirchhoff–Love shell: From design to shape optimization of non-conforming stiffened multipatch structures, *Computer Methods in Applied Mechanics and Engineering* 349 (2019) 774–797.
- [12] H. Zhao, D. Kamensky, J. T. Hwang, A. J. Herrema, Shape optimization of non-matching isogeometric shells with moving intersections, *Computer Methods in Applied Mechanics and Engineering* (2024).
- [13] T. Hirschler, R. Bouclier, A. Duval, T. Elguedj, J. Morlier, Isogeometric modeling for the optimal design of aerostructures, *AIAA Journal* (2019).
- [14] H. Zhao, D. Kamensky, J. T. Hwang, A. J. Herrema, Open-source shape optimization for isogeometric shells using FEniCS and OpenMDAO, *arXiv preprint arXiv:2410.02225* (2024).
- [15] K. Y. Sze, X. H. Liu, S. H. Lo, Popular benchmark problems for geometric nonlinear analysis of shells, *Finite Elements in Analysis and Design* 40 (11) (2004) 1551–1569. doi:10.1016/j.fine1.2003.11.001.
- [16] COMSOL, Shape optimization of a shell, COMSOL Multiphysics Application Library, Optimization Module, model `shell_shape_optimization`, COMSOL Multiphysics v6.0, accessed July 2026 (2021).
- [17] P. M. Naghdi, The theory of shells and plates, *Handbuch der Physik*, Vol. VIa/2 (1973) 425–640.
- [18] D. Chapelle, K.-J. Bathe, *The Finite Element Analysis of Shells—Fundamentals*, 2nd Edition, Springer, 2011.
- [19] C. Lovadina, F. Auricchio, Energy norm a posteriori error estimates for mixed finite element methods, *Computer Methods in Applied Mechanics and Engineering* Partial selective reduced integration analysis reference (2005).
- [20] M. C. Delfour, J.-P. Zolésio, *Shapes and Geometries: Metrics, Analysis, Differential Calculus, and Optimization*, 2nd Edition, SIAM, 2011.
- [21] D. A. Ham, L. Mitchell, A. Paganini, F. Wechsung, Automated shape differentiation in the unified form language, *Structural and multidisciplinary optimization* 60 (5) (2019) 1813–1820.
- [22] V. H. Schulz, M. Siebenborn, K. Welker, Efficient pde constrained shape optimization based on Steklov–Poincaré-type metrics, *SIAM Journal on Optimization* 26 (4) (2016) 2800–2819. doi:10.1137/15M1029369.
- [23] B. S. Lazarov, O. Sigmund, Filters in topology optimization based on Helmholtz-type differential equations, *International Journal for Numerical Methods in Engineering* 86 (6) (2011) 765–781. doi:10.1002/nme.3072.
- [24] M. Hojjat, E. Stavropoulou, K.-U. Bletzinger, The vertex morphing method for node-based shape optimization, *Computer Methods in Applied Mechanics and Engineering* 268 (2014) 494–513. doi:10.1016/j.cma.2013.10.015.
- [25] D. Ridzal, D. P. Kouri, ROL: Rapid Optimization Library, Tech. rep., Sandia National Laboratories, trilinos package (2014).